Introduction

Class 1 of 24: Natural Language Processing

CSE 4811 / DS 4213 (Fall 2026)

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Two kinds of language

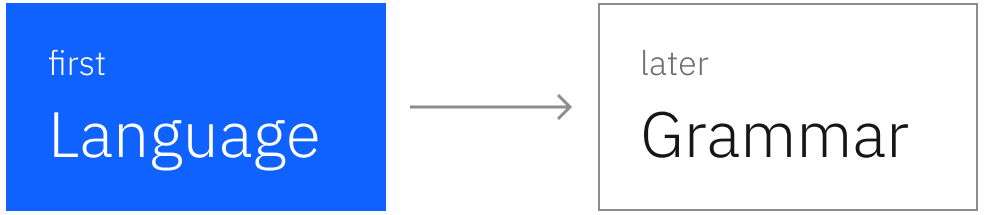
What is a **natural** language?

বাংলা
Bangla

English
English

العربية
Arabic

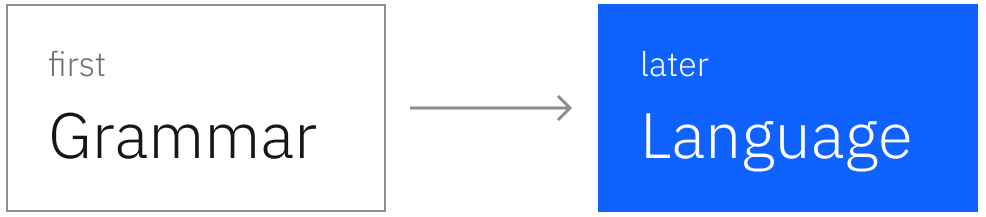
Français
French



Then what is an **unnatural** language?

Formal

the opposite of natural



Four formal languages

Algebra

$$(a+b)^2 = a^2 + 2ab + b^2$$

Logic

$$\forall x \left(\text{Student}(x) \rightarrow \exists y \, \text{Enrolled}(x, y)\right)$$

Legal contract

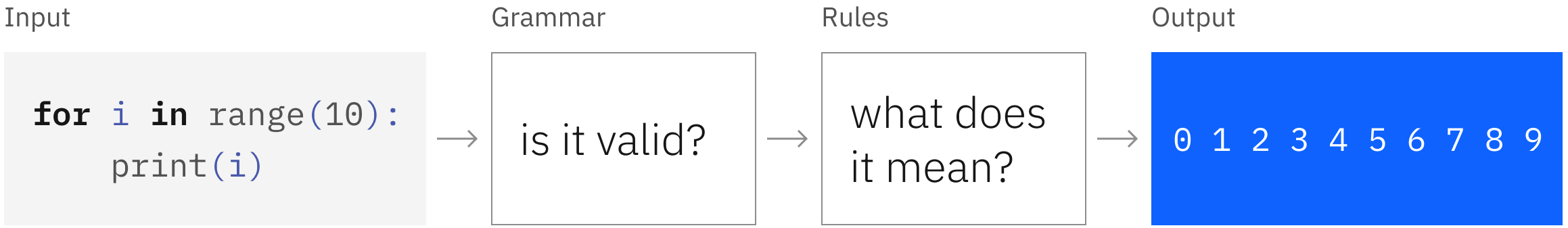
defined terms
numbered clauses
no pronouns

Code

```
for i in range(10): print(i)
```

Each symbol has **one meaning**.

Processing a formal language is easy



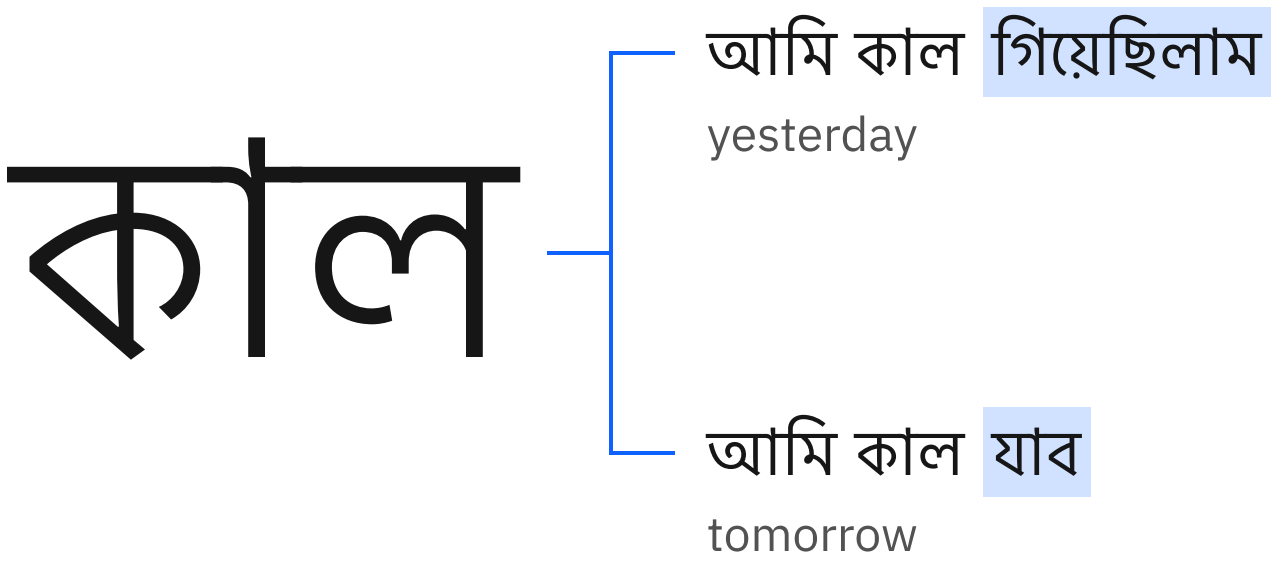
Solved by the 1960s. Not in this course (Compiler Design).

NLP: text in, useful output out

Task	Input	Output	Output type
News topic classification today	`Apple security flaw could allow hackers to intercept emails`	→ science and technology	one label
Sentiment analysis	`The food was cold and the service slow.`	→ `negative`	one label
Part-of-speech tagging	`Dhaka beat India`	→ `noun` `verb` `noun`	one label per word
Translation, English to Bangla	`The sun is rising`	→ সূর্য উঠছে	a sentence
Query to a database	`Find all students in section A`	→ `Rafi and Shondha`	a query, then its answer

Every output can be written as **text**.

Why natural language is hard



English

`I saw her duck.`

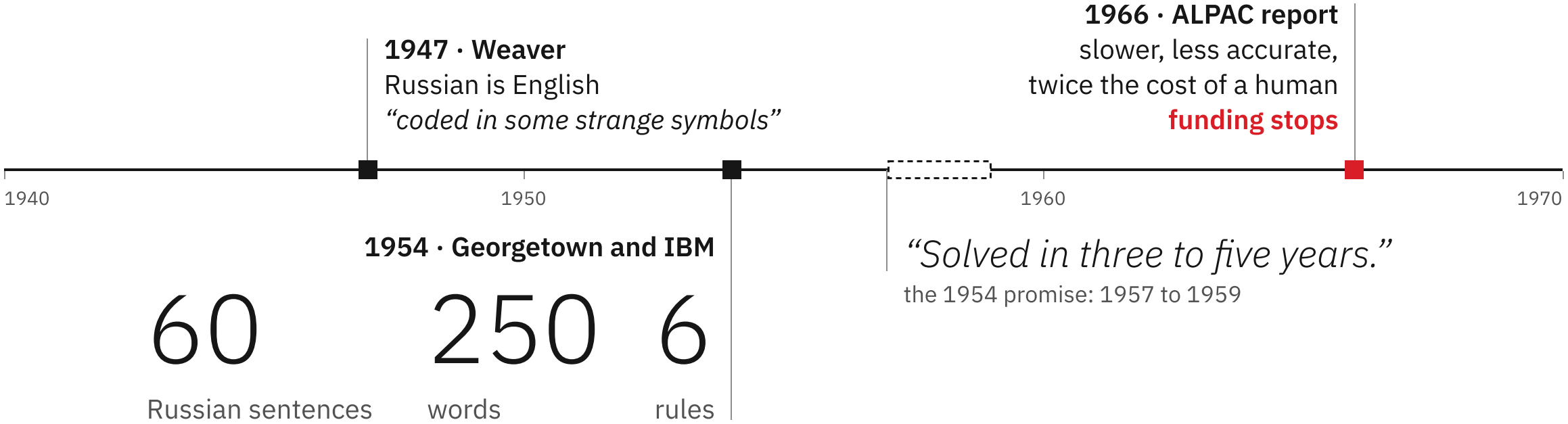
Ambiguity
one word, many meanings

World knowledge
“The box was in the pen.”
Bar-Hillel, 1960

Variation
spelling, dialect, new words

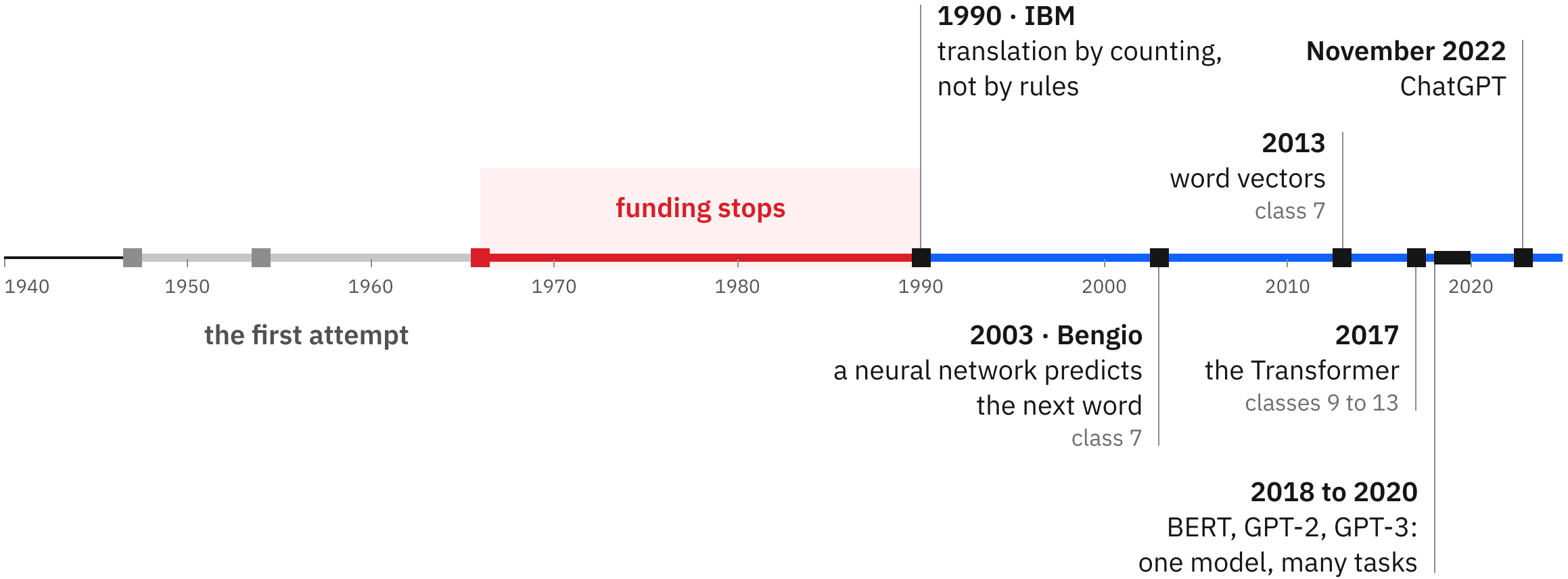
No complete grammar
nobody has written all the rules

The first attempt: rules, 1947 to 1966



Rules cannot handle **scale and ambiguity**.

From rules to learning



One idea: **stop writing rules; learn from text.**

Forty years, then two months

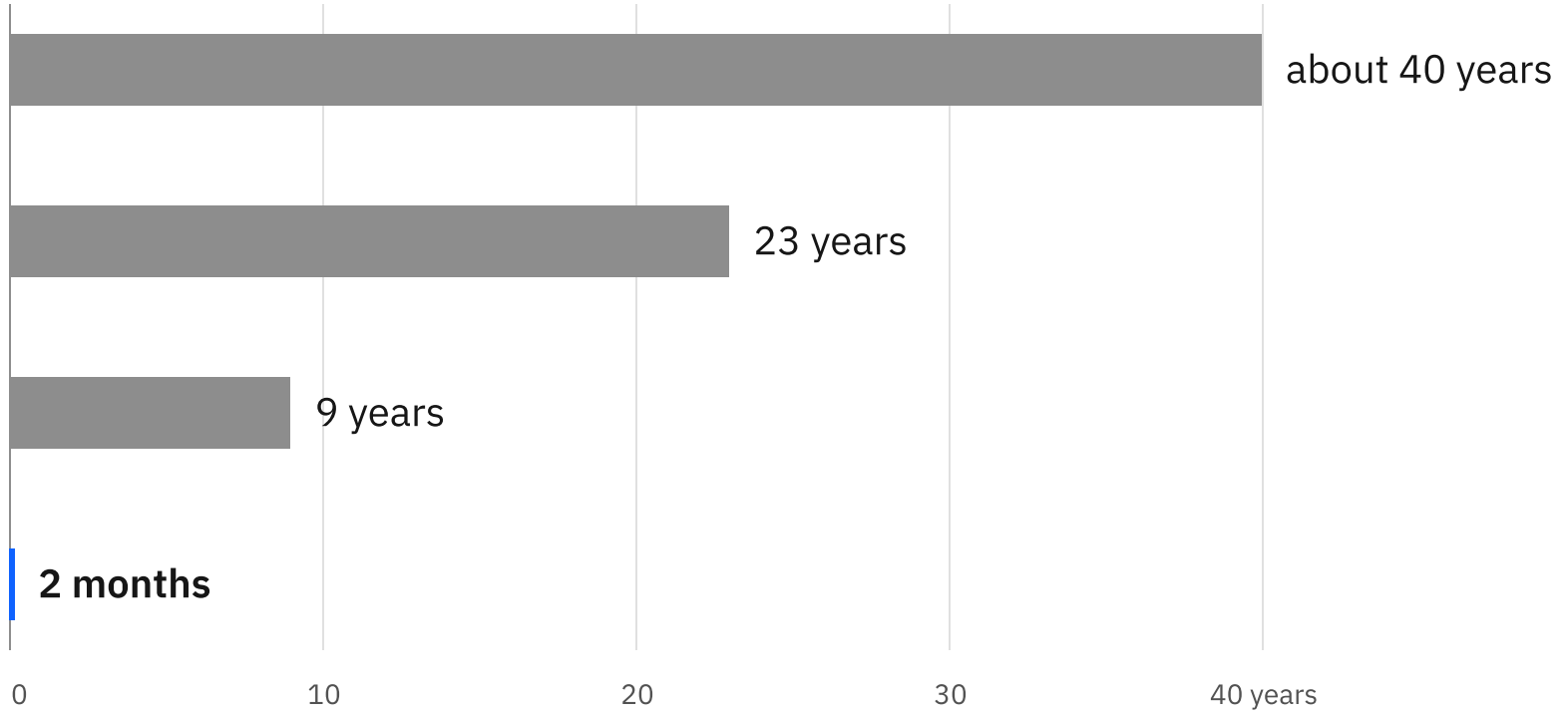
Time from launch to mass use

Electricity
1882 → half of US homes, about 1925

Personal computer
1977 → half of US homes, 2000

Internet
1991 → half of US adults, 2000

ChatGPT
Nov 2022 → 100 million users, Jan 2023

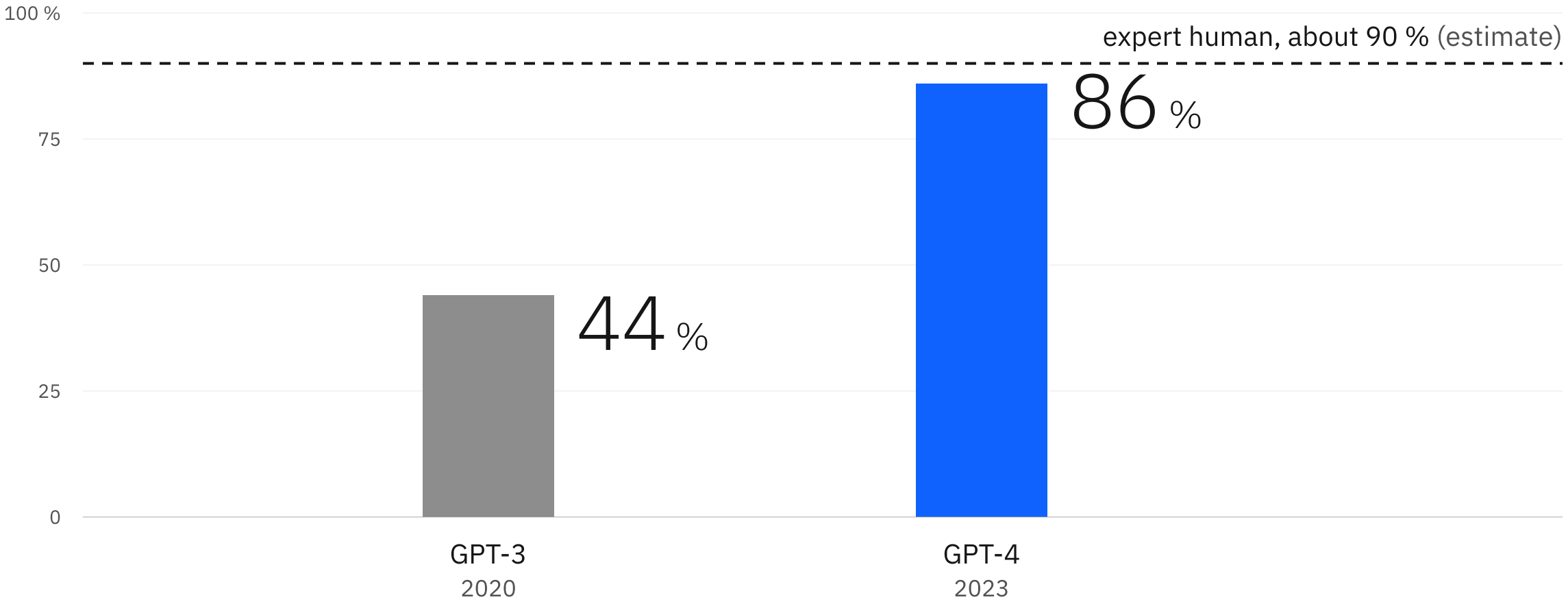


800 million

weekly users · OpenAI, October 2025

Failing student to near expert in three years

MMLU: exam questions from 57 university subjects



AI assistants at work: three studies, 2023

Controlled studies: the same work, with and without an AI assistant

40%

less time on writing tasks; quality rated 18 % higher

Noy and Zhang, *Science*

14%

more issues resolved per hour by 5,000 support agents; 34 % for new staff

Brynjolfsson, Li and Raymond

40%

higher quality from 758 consultants, 25 % faster

Dell’Acqua and others

Today’s task

A news item in, one of six topics out. In Bangla and in English.

- national politics
- international
- sports
- business and economy
- entertainment
- science and technology

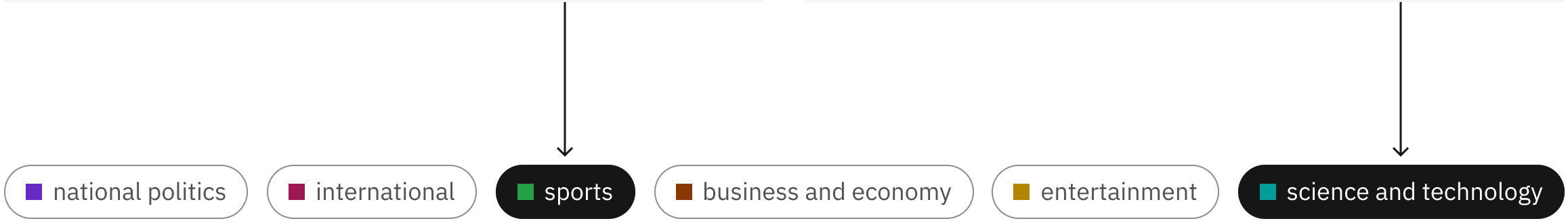
Two items from the training set

Bangla

এনামুলের ফিফটিতে ঢাকার সংগ্রহ ১৮২ এনামুল হক বিজয়ের ফিফটিতে ঢাকা প্লাটুন ১৮২ রানের চ্যালেঞ্জিং স্কোর গড়েছে। ...

English

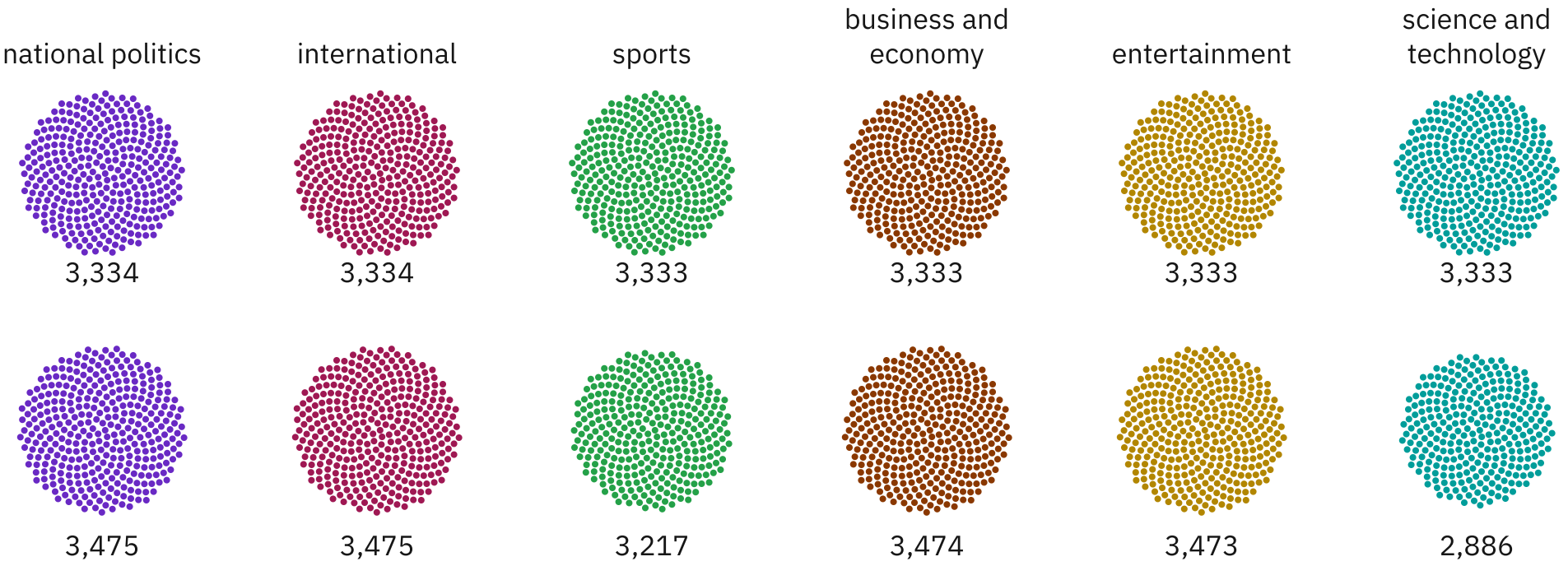
Apple Security Flaw Could Allow Hackers To Intercept Emails. Apple released software patches and an update …



How can a program do this?

Our dataset: six topics, two languages

Training set
one dot = ten items



44,000
labeled items



+ a hidden test set on Kaggle

Metric: macro-F1



random ≈ 0.17 perfect = 1.0

Bangla: Potrika, Ahmad et al. 2022, CC BY 4.0. English: News Category Dataset, Misra 2022, HuffPost 2012 to 2022, CC BY 4.0. Both modified: extracts only, labels merged.

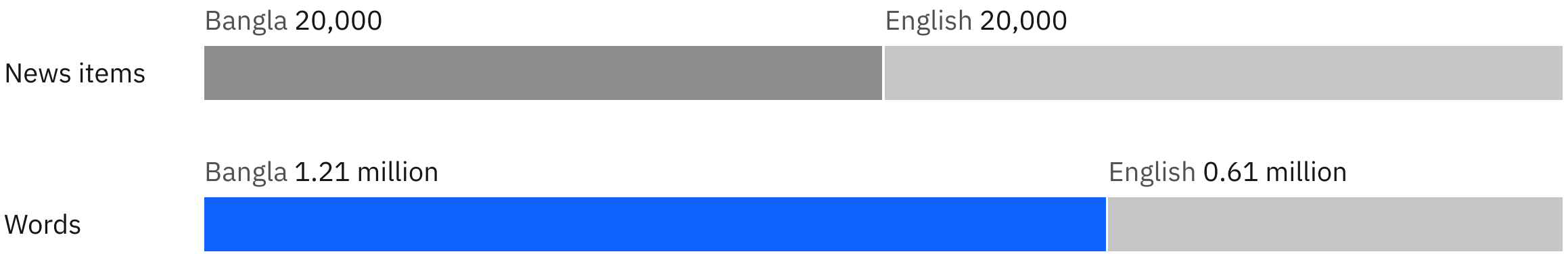
Corpus: the training data

A body of text collected for study, of known origin: author, date, language, purpose

40,000
news items

1.82 million
words

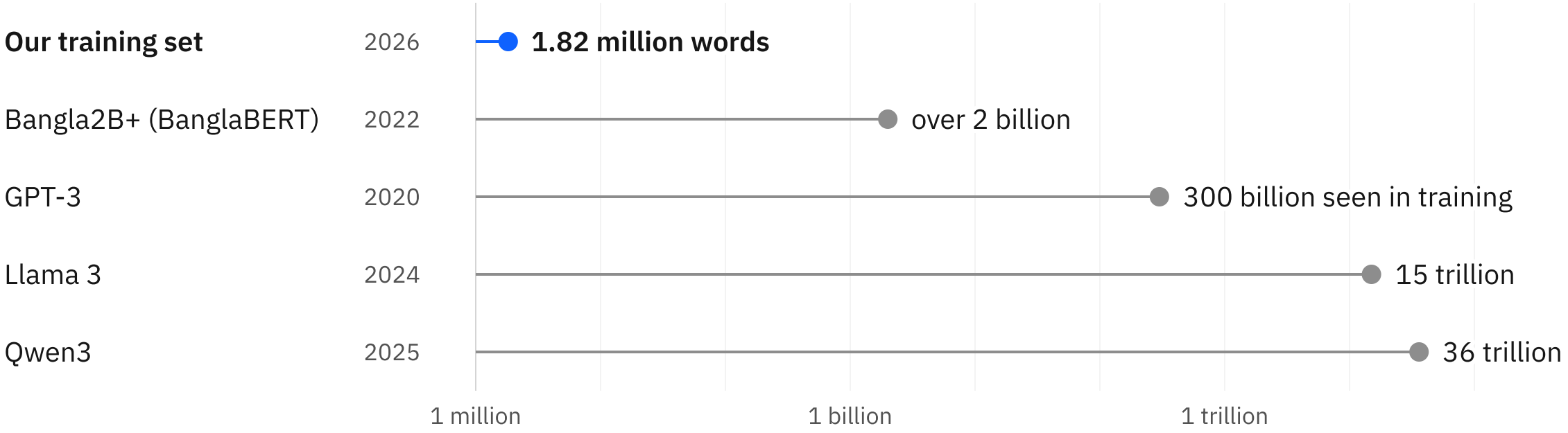
11.8 million
characters



A Bangla item has about twice as many words as an English item.

How big are training corpora?

Size in tokens, log scale (×10 per gridline). Ours is counted in words.



WebText (GPT-2)	2019	40 GB; tokens not published
Claude; GPT-4 and later		not published

What is a token?

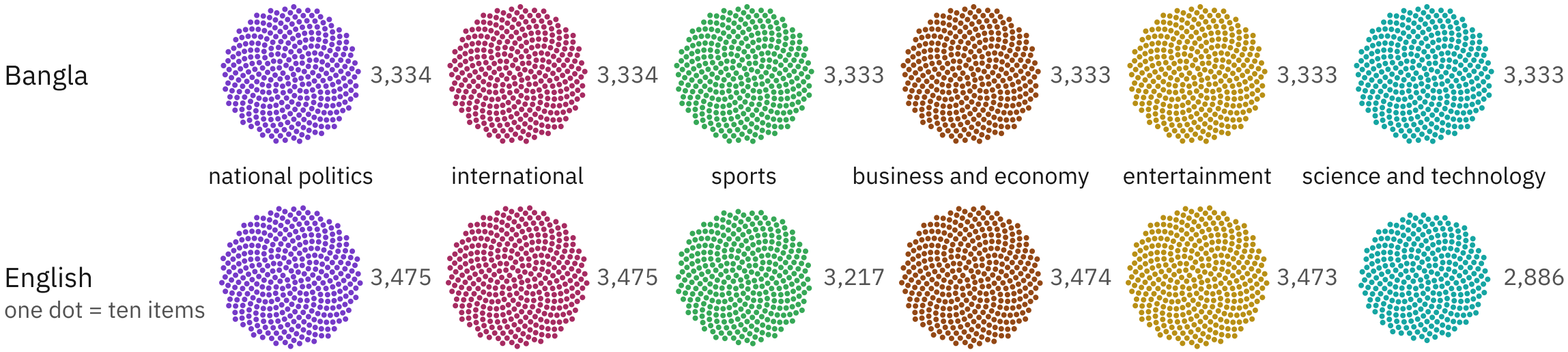
Tokenizer: text → tokens

Text	ভারতকে	newer	Distinct units	Sequence
Characters	ভ ◌া র ত ক ে◌	n e w e r	a few hundred	long
Subwords	ভারত | কে	new | er	32,000 to 262,000 chosen	between
Words today	ভারতকে	newer	tens of thousands and growing	short

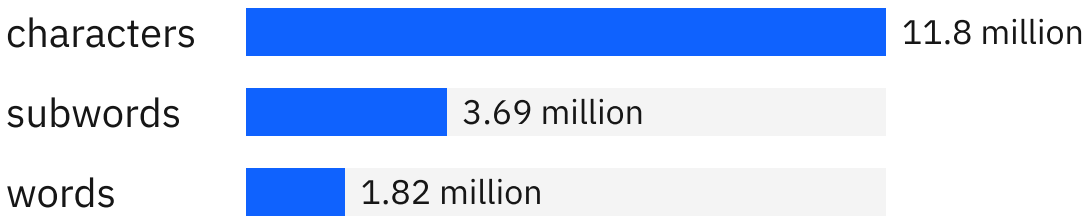
Today: words, cut by one regular expression.
Class 4: subwords, learned from the corpus (byte-pair encoding).

Three ways to cut text into tokens

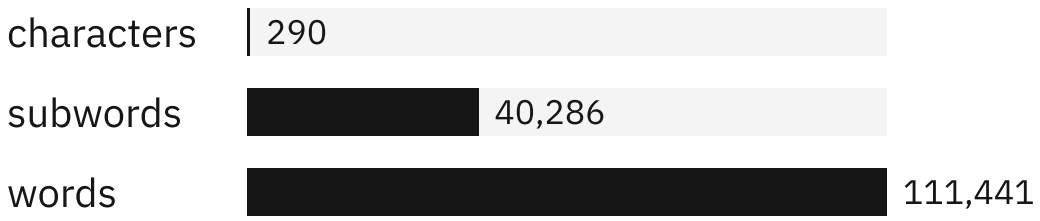
Three tokenizations of the news topic dataset



Tokens in the training set



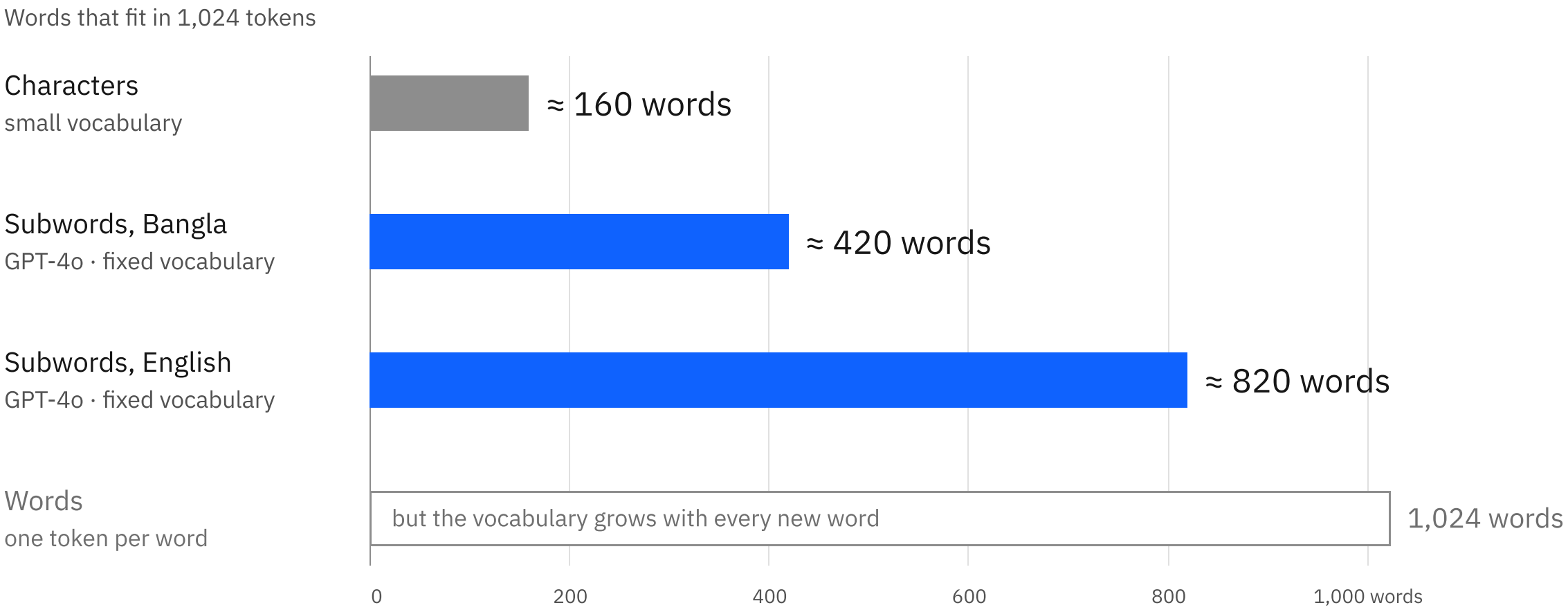
Vocabulary of the training set



subwords: the o200k_base tokenizer, used in GPT-4o

The context window

A model reads a fixed number of tokens at a time. GPT-2: 1,024 tokens.



How big is a vocabulary?

Our training set, words

1.82 million

word instances

running words, repeats counted

111,441

word types

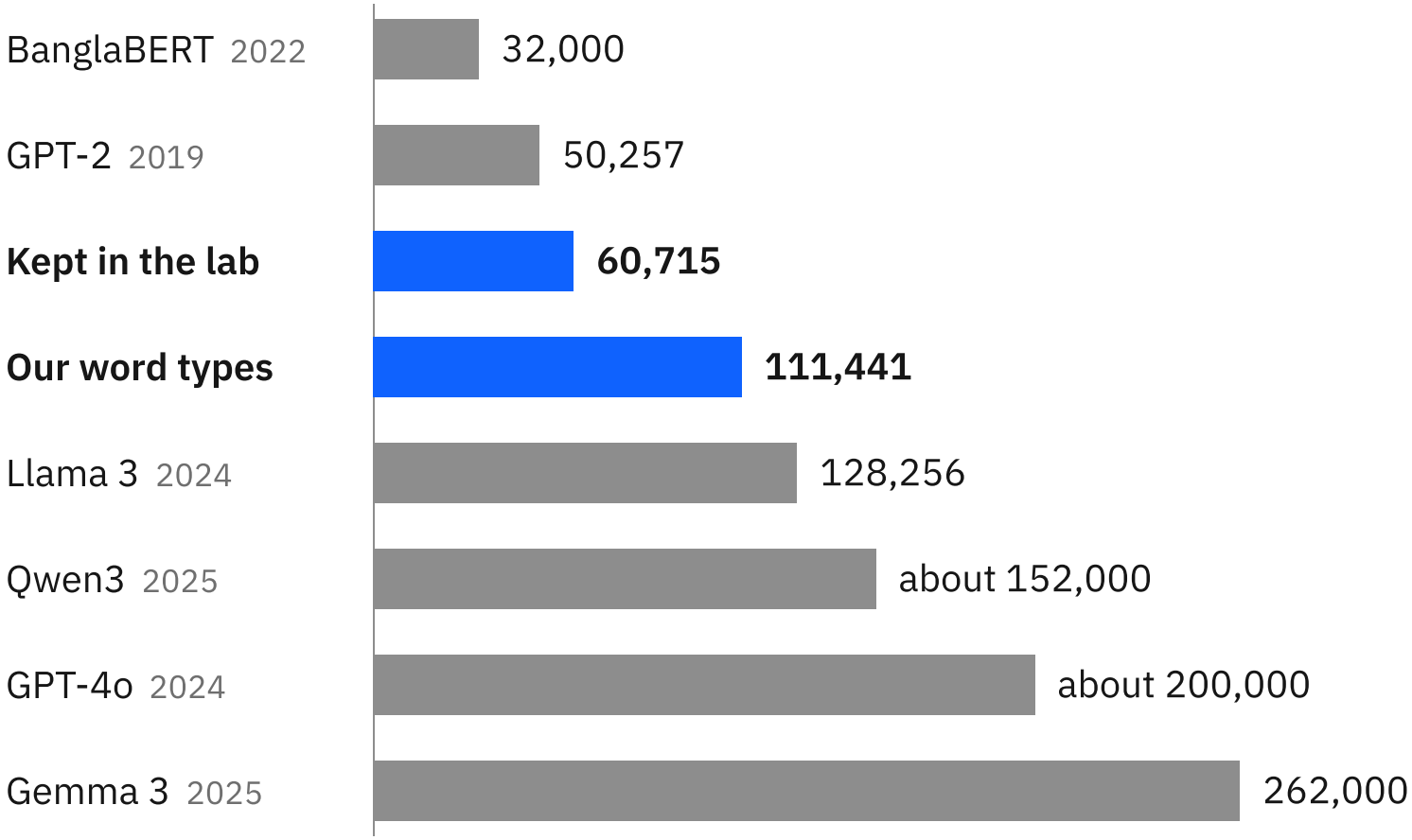
distinct words: today's vocabulary

60,715

kept in the lab

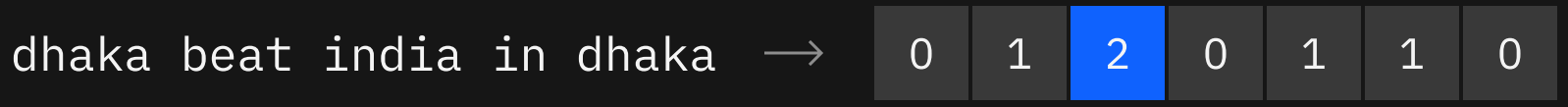
types occurring ≥ 2 times

Vocabulary size: subword tokenizers and our words



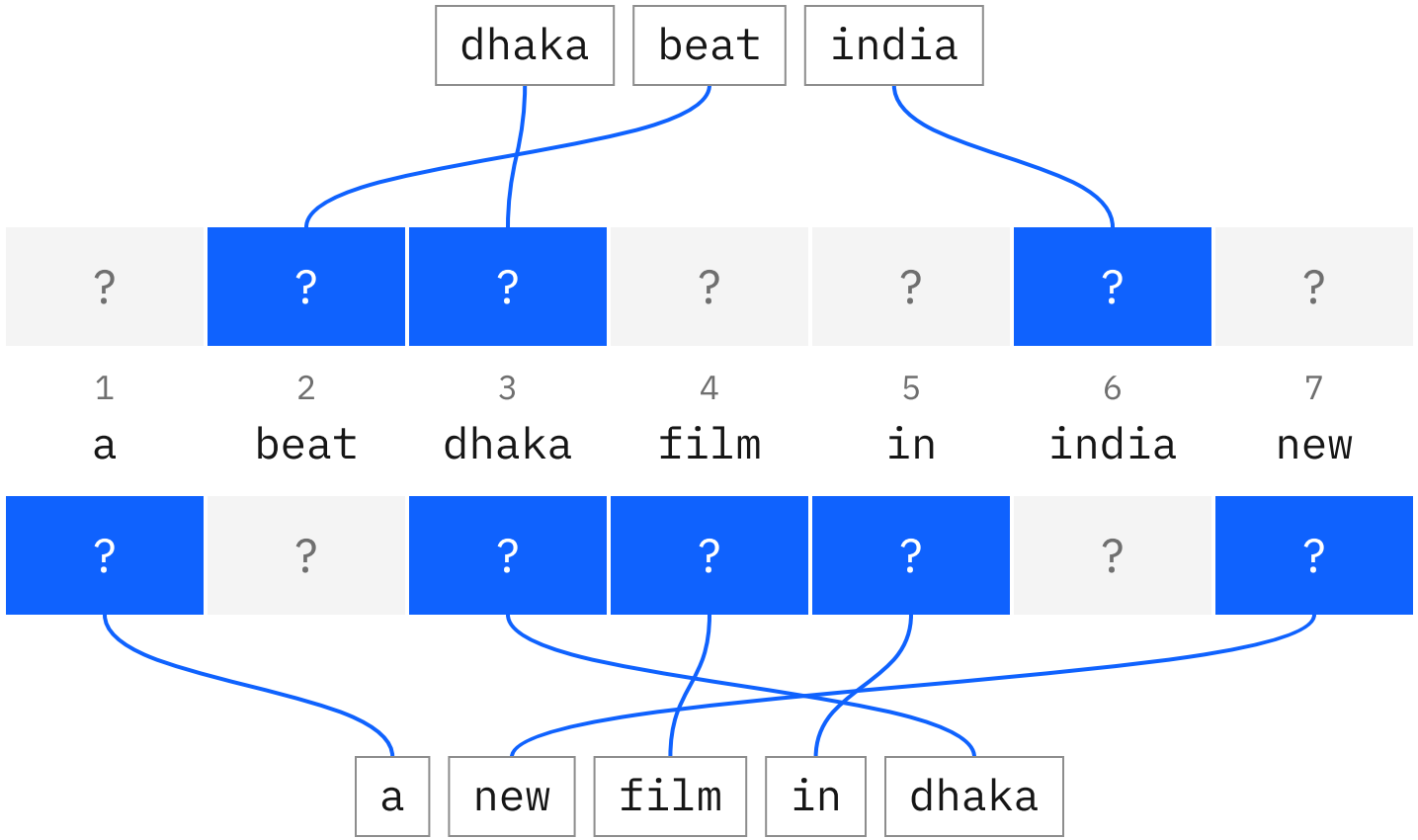
From text to vectors

A text of any length becomes one vector of numbers.



Each text becomes a vector

One position per vocabulary word: $x \in \mathbb{R}^{|V|}$



Our corpus

111,441

word types, Bangla and English

$x \in \mathbb{R}^{111,441}$

5 of 7 positions used

length is always 7

What value goes in each position?

Bag of words: count every word

The simplest representation: count each vocabulary word in each text

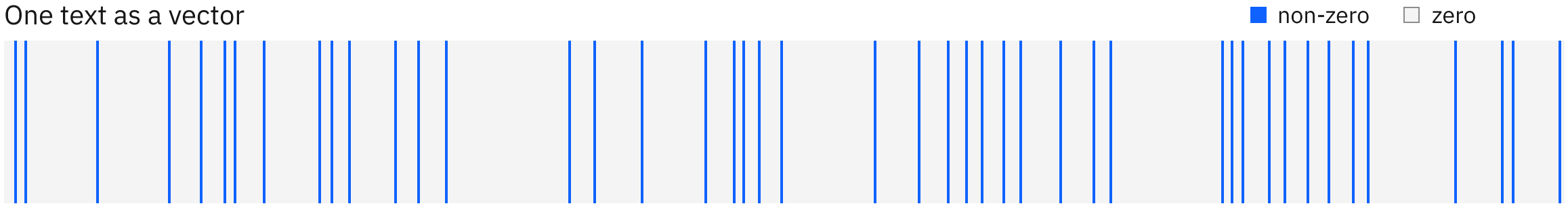
Three toy texts	a	beat	dhaka	film	in	india	new
dhaka beat india in dhaka	0	1	2	0	1	1	0
india beat dhaka in dhaka	0	1	2	0	1	1	0
a new film in dhaka	1	0	1	1	1	0	1

Rows: texts. Columns: the vocabulary. Each cell: how often the word occurs in the text.

Rows 1 and 2 are the same: **word order is lost**. The largest count, dhaka = 2, is not a topic word.

Bag of words: almost all zeros

One text as a vector



111,441 columns, about 45 non-zero per text. **Store only the non-zeros.**

Word types in the training set 111,441



60,715 kept
seen at least twice

50,726 dropped
seen once: no use on new text

Two problems of raw counts

A raw count is also called the **term frequency (tf)**

Length

	beat	dhaka	india	row length
dhaka beat india	1	1	1	→
the same, twice	2	2	2	→
both, scaled	0.58	0.58	0.58	→ 1

Common words

	a	beat	dhaka	film	in	india	new
1	0	1	2	0	1	1	0
2	0	1	2	0	1	1	0
3	1	0	1	1	1	0	1

dhaka, in: in all three texts

dhaka = 2: the largest count, not a topic word

Solution

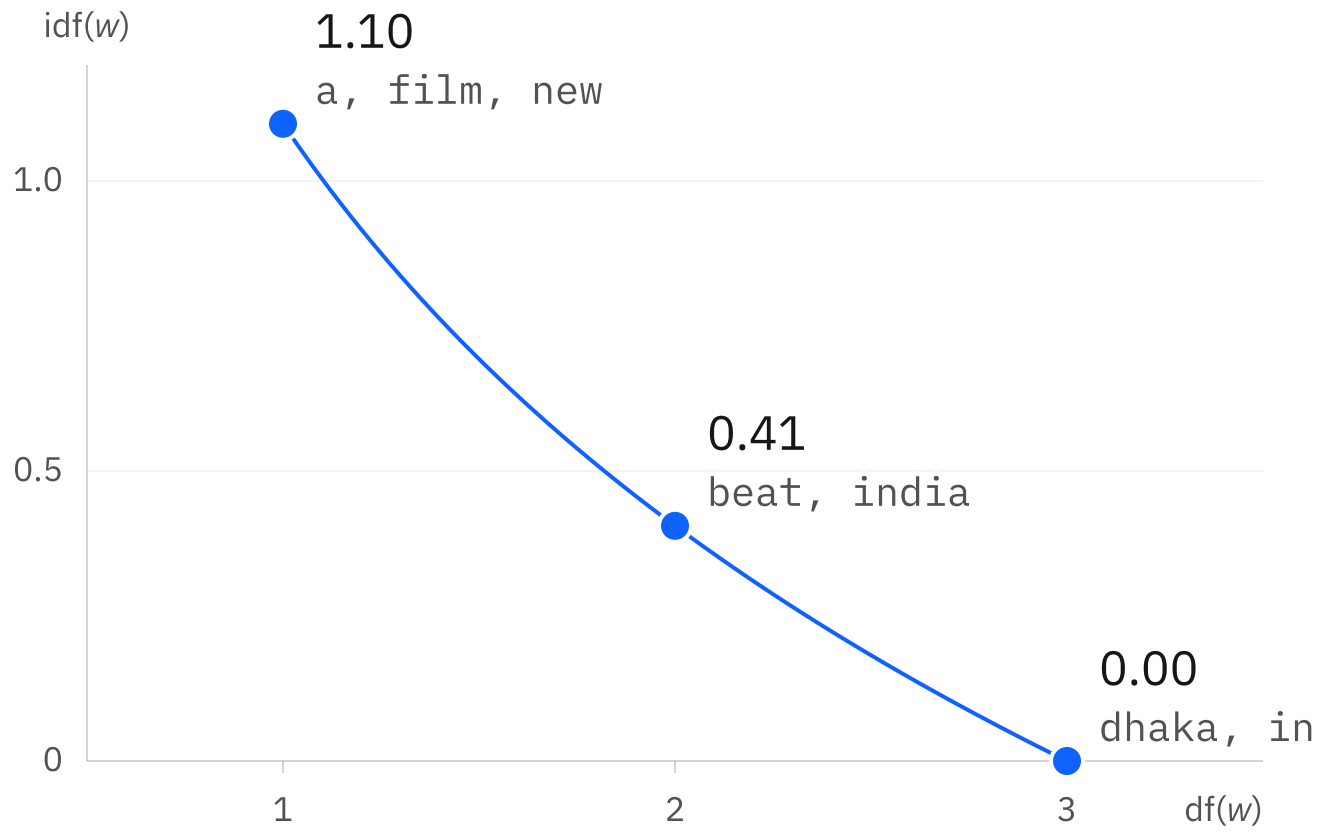
Scale each row to length 1

Solution

Weight each word by how rare it is: idf

Inverse document frequency

N: number of texts. $\mathrm{df}(w)$: number of texts that contain w.



$$\mathrm{idf}(w) = \log \frac{N}{\mathrm{df}(w)}$$

The three toy texts, $N = 3$

1. `dhaka beat india in dhaka`
2. `india beat dhaka in dhaka`
3. `a new film in dhaka`

In all texts → 0. In one text → the highest weight.

From counts to TF-IDF

✓ Counts (tf) ✓ × idf ✓ ÷ row length

		a	beat	dhaka	film	in	india	new	row length
× idf		1.10	0.41	0.00	1.10	0.00	0.41	1.10	
1	dhaka beat india in dhaka	0	0.71	0	0	0	0.71	0	1
2	india beat dhaka in dhaka	0	0.71	0	0	0	0.71	0	1
3	a new film in dhaka	0.58	0	0	0.58	0	0	0.58	1

dhaka, in → 0; rarer words keep their weight; every row has length 1.

TF-IDF on the real corpus

idf on the 40,000 training texts

Which is largest?

common words



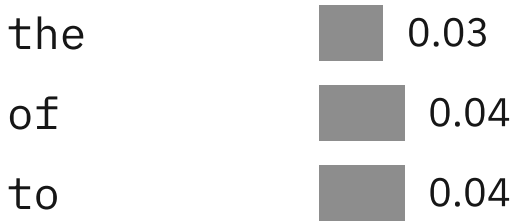
topic words



One news item (headline shown; weights from the whole item)

“Apple Security Flaw Could Allow Hackers To Intercept Emails”

lightest



heaviest



Topic words get the highest weights.

Cosine similarity: the angle between two texts

$$\cos\theta = \frac{x \cdot y}{\|x\| \, \|y\|} = x \cdot y$$

Every row has length 1.

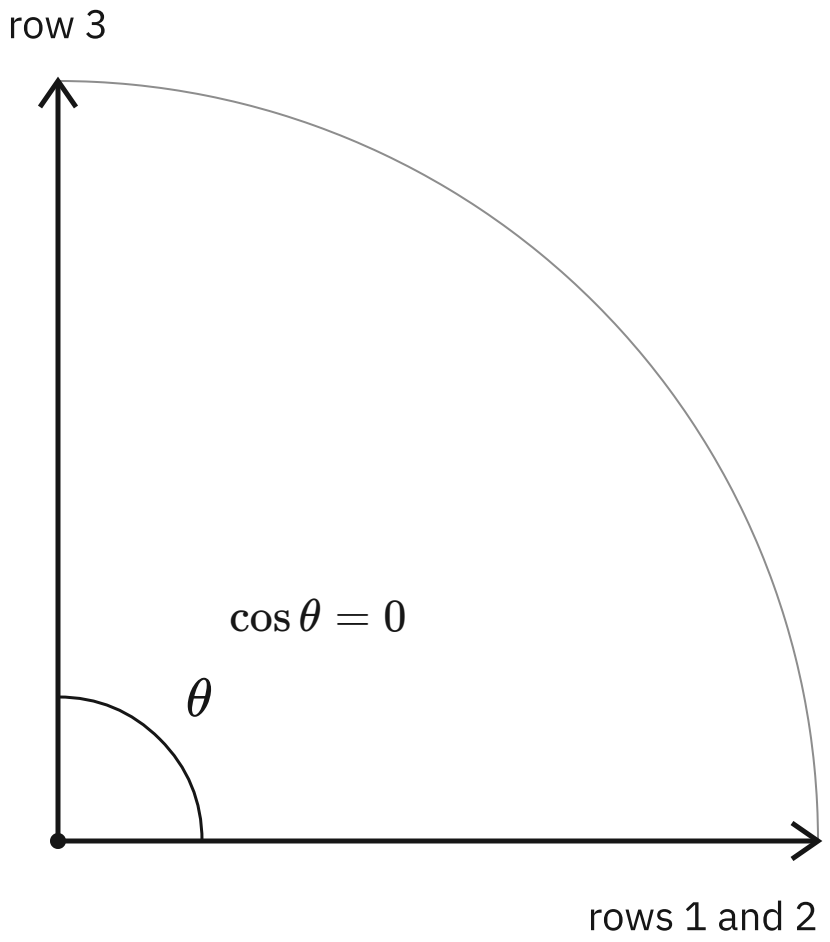
1 same words, same proportions

0 no shared word with weight above 0

Toy texts

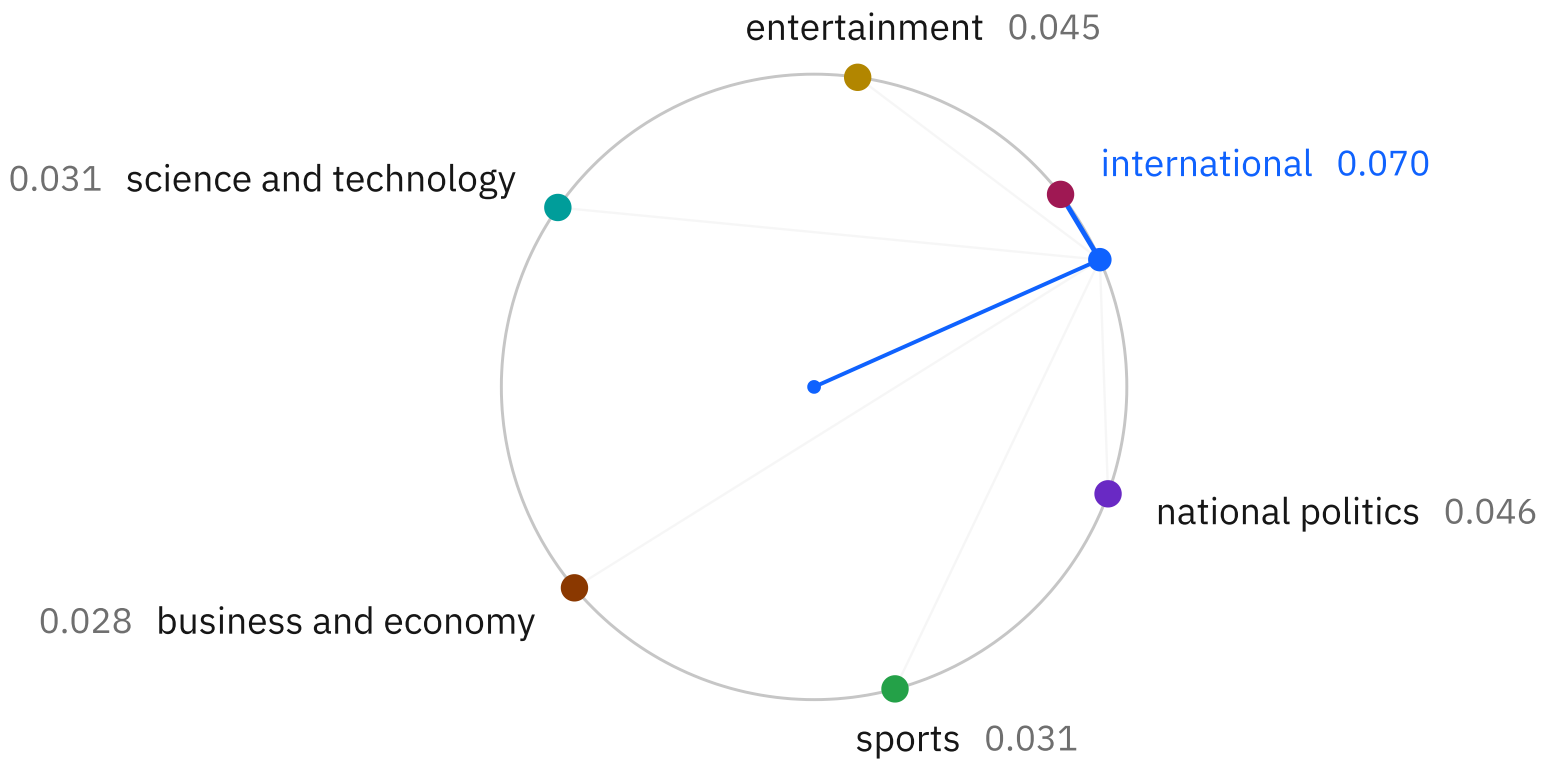
1.00 rows 1 and 2

0.00 rows 1 and 3



Classify by the nearest topic average

1 2 3 4 **Predict.** The topic with the highest cosine similarity is our prediction.



international

চীনের হংকং-উইঘুর নীতির সমালোচনায় ট্রুডো সময়ের আলো ডেস্ক : চীনের হংকং-উইঘুর নীতির কঠোর সমালোচনা করেছেন কানাডার প্রধানমন্ত্রী জাস্টিন ট্রুডো। তিনি বলেছেন, হংকংয়ে চীনা দমননীতি এবং উইঘুর মুসলিমদের আটকে রাখার পাল্টা প্রতিক্রিয়া

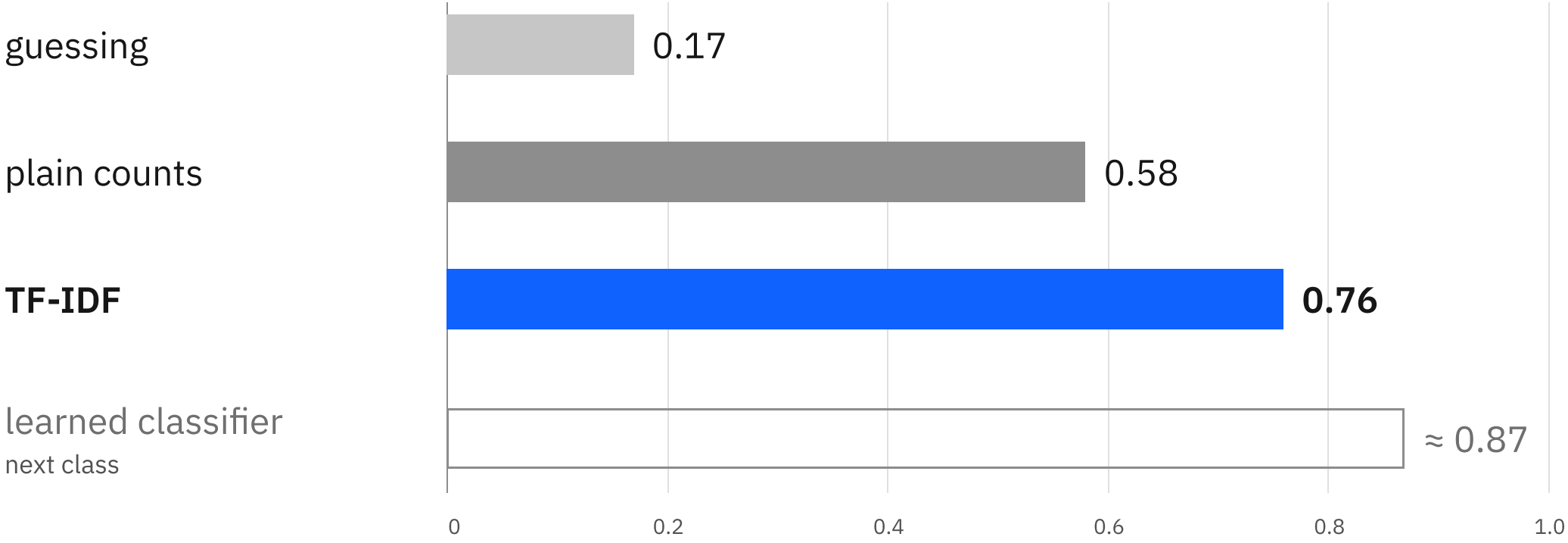
59 of 62 words known

international	0.070
national politics	0.046
entertainment	0.045
sports	0.031
science and technology	0.031
business and economy	0.028

A sketch: each text is a vector with 111,441 positions. The six cosines are computed for real.

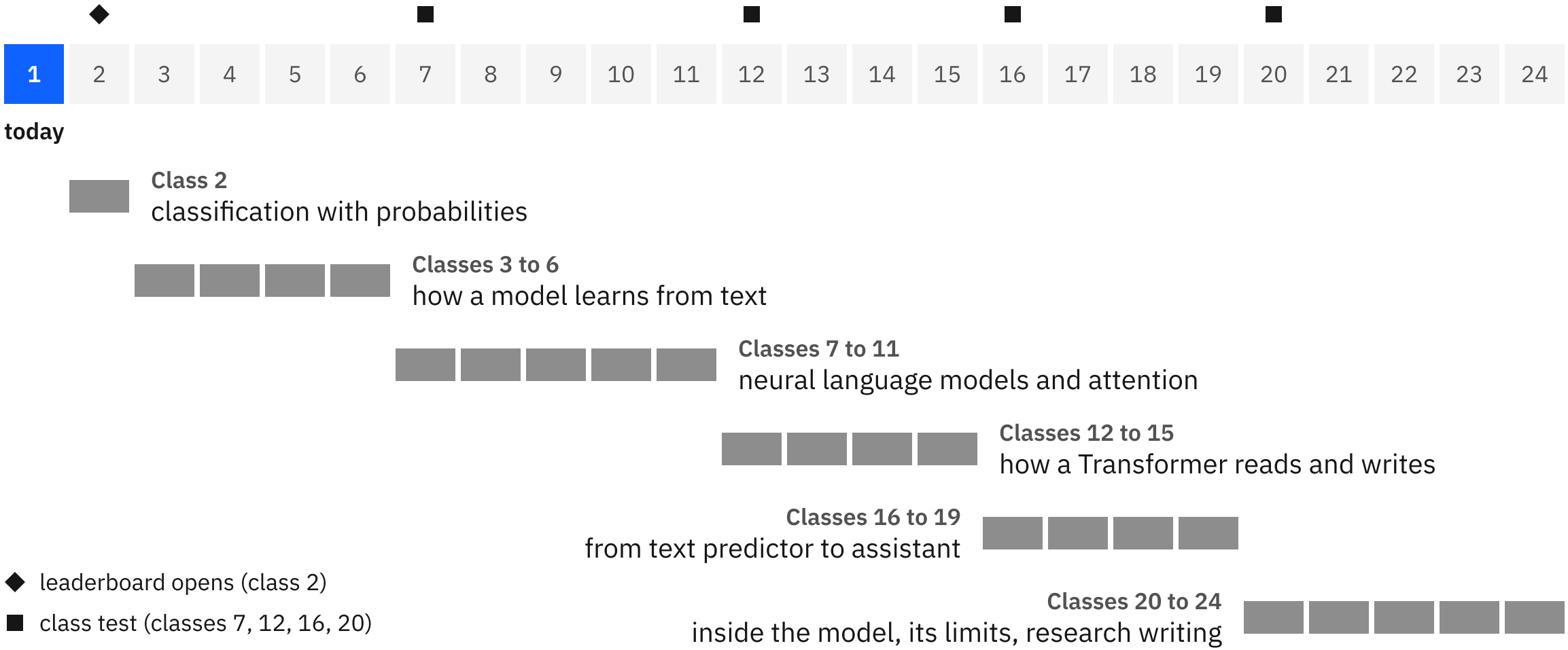
Does TF-IDF matter?

The same classifier on the 4,000 development items. Macro-F1: perfect 1.0, random about 0.17.



0.58 → 0.76 macro-F1, counts → TF-IDF

The course ahead



Assessment: 100 marks

attendance

5

leaderboard

10

final examination

40



20

four class tests

classes 7, 12, 16, 20

25

term paper

conference format, peer review

Class tests

Pen and paper, 25 minutes, start of class.

Leaderboard

Weekly Kaggle rounds, all sections together. Opens in class 2.

Details

The outline on Moodle.

Today in one picture

Language	A natural language grew among people; a formal one has its grammar first.
History	Rules failed by 1966; learning from text took over.

News item
input

```
dhaka beat
india in
dhaka
```

Words
tokens

dhaka | beat | india | in | dhaka

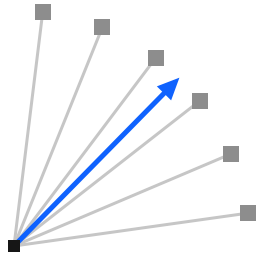
Bag of words
counts

dhaka	2
beat	1
india	1
in	1

TF-IDF
weights

dhaka	0
beat	0.71
india	0.71
in	0

Cosine
to six topic averages



Topic
the largest cosine

- national politics
- international
- sports
- business and economy
- entertainment
- science and technology

Macro-F1
4,000 dev items

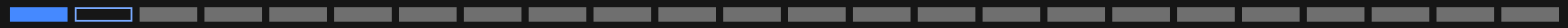
0.58 → 0.76

with counts → with TF-IDF

Thank you.

Next — Class 2: classification with probabilities

Reading — Jurafsky and Martin, chapter 4



References

Textbook

Jurafsky, D. and Martin, J. H. (2025). *Speech and Language Processing*, 3rd ed. draft. Section 2.1, p. 34 (types and instances) and p. 36 (Heaps' law); section 2.4, pp. 42–46 (tokens, subwords, vocabulary sizes); section 2.5, p. 47 (corpora, datasheets, English bias); chapter 4 (reading for class 2).

Datasets

Ahmad, I., Al Qurashi, F., Abozinadah, E. and Mehmood, R. (2022). *Potrika dataset.* Mendeley Data, doi:10.17632/v362rp78dc.4, CC BY 4.0. Modified: headline and first 60 words only, labels merged.

Misra, R. (2022). *News Category Dataset.* Kaggle, CC BY 4.0; arXiv:2209.11429. HuffPost, 2012–2022. Modified: six categories kept and merged.

Work studies

Noy, S. and Zhang, W. (2023). Science 381.

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